

RITWIK BISWAS

Software Engineer | Backend, Distributed Systems & AI

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EDUCATION

Indian Institute of Technology Kharagpur (IIT KGP)
Dual Degree (B.Tech + M.Tech) in Mechanical Engineering | CGPA: 7.84/10

2021 - 2026

EXPERIENCE

Taskly Technologies Inc. | Software Engineering Intern

Remote, Canada | Mar 2025 - Jul 2025

- Shipped core JobSense modules across Next.js, FastAPI and PostgreSQL, delivering five AI-assisted workflows for JD analysis, resume scoring, ATS checks, interview preparation and company research through integrated frontend, API and database flows.
- Integrated Groq and Firebase Authentication, enforcing JWT-secured user sessions across protected JobSense AI endpoints.
- Resolved JavaScript-rendered scraping failures, API payload bottlenecks, response-filtering errors, environment misconfigurations and Docker conflicts; containerized the affected services to stabilize API behavior and support reliable Vercel releases.

Search-in | Full-Stack Developer Intern

Remote, Bengaluru | Jun 2024 - Jul 2024

- Delivered a React admin dashboard and Node.js/Express APIs for 10K+ users, 1K+ orders and 500+ managed records; automated analytics, order management and operational workflows, increasing engagement by 20% while supporting order and record operations.
- Optimized MongoDB access across high-traffic operations using indexes, Redis caching and query tuning, reducing database latency and improving request throughput for concurrent analytics, order and record-management workloads across the platform.
- Streamlined the end-to-end order workflow to reduce processing time by 30%, then deployed frontend assets to AWS S3 and cloud-hosted services, improving availability and maintainability across React admin, order-management and operational workflows.

Pepcorns | Full-Stack Developer Intern

Remote, Pune | Mar 2024 - Apr 2024

- Built a Node.js/SQL referral-rewards system across three workflows - attribution, reward issuance and social sharing - with end-to-end tracking; integrated reusable React components with backend APIs and database operations, reducing duplication across user interfaces.

PROJECTS

ThesisLens | AI-Powered Investment Research & Intelligence Platform | GitHub

Sep 2026 - Present

Python | FastAPI | Azure OpenAI | Azure AI Search | PostgreSQL | RAG | BM25 | SentenceTransformers | Docker | GitHub Actions

- Architected an evidence-grounded investment research platform that transforms financial filings through semantic chunking, Azure OpenAI embeddings and Azure AI Search, while extracting validated financial KPIs into PostgreSQL for structured analysis.
- Engineered deterministic SQL / RAG / combined routing with stateful report research and claim-level citation validation; built an investment-thesis engine for grounded bull/bear cases, catalysts, risks and management outlook, rejecting unsupported claims.
- Built a 44-query, 1,012-pair content-adjudicated retrieval benchmark with Recall@K, Precision@K, MRR and nDCG; fine-tuned a local MiniLM cross-encoder, improving holdout nDCG@5 from 0.704 to 0.811 and Recall@3 from 0.448 to 0.604.
- Productionized the service with request-level observability, Docker, liveness/readiness probes and zero-cloud GitHub Actions CI; retained lexical retrieval after the reranker added ~3.19 s/query CPU latency and reduced top-rank MRR.

TraceForge | Distributed AI Agent Reliability & Replay Platform | GitHub

Jul 2026 - Present

Go | Kafka | OpenTelemetry | Python | LangGraph | PostgreSQL | Docker | Kubernetes | Terraform | Distributed Systems

- Architected a non-blocking distributed telemetry ingestion pipeline around a resilient Go gateway with versioned capture contracts, schema validation, idempotent event processing and Kafka-backed asynchronous delivery; decoupled latency-sensitive AI-agent execution from downstream processing to enable trace reconstruction, evaluation and replay across LangGraph and external API workflows.
- Engineered the foundation for deterministic AI-agent replay and regression testing by capturing execution context and freezing nondeterministic LLM/tool boundaries, enabling LangGraph and external API workflows to be reproduced without depending on changing live responses; hardened ingestion through configuration, validation, publisher-failure and distributed failure-path testing.
- Extending TraceForge into a production-grade agent reliability platform with OpenTelemetry-compatible tracing, PostgreSQL-backed trace/span persistence, asynchronous evaluators, replay-fixture generation, original-vs-replay diffs, API-key isolation, consumer-lag/throughput/latency observability, containerized workers and Kubernetes deployment with Terraform-managed infrastructure.

Converge | Distributed Real-Time Collaboration Platform | GitHub

Aug 2026 - Present

TypeScript | React | Konva | Fastify | Socket.IO | PostgreSQL | Redis | IndexedDB | Docker | Playwright | k6

- Architected a horizontally scalable, server-authoritative collaboration engine using PostgreSQL advisory-lock sequencing and deterministic reducers; combined versioned protocols, idempotent commands, transactional operation/projection/outbox/receipt commits, atomic change sets and Redis-backed multi-replica WebSocket fan-out to preserve consistent state across distributed clients.
- Engineered offline-first synchronization, concurrency control and failure recovery using IndexedDB persistence-before-optimism, durable command queues, generation-fenced reconnects, fixed-watermark catch-up, snapshot + operation-tail replay, canonical state hashing, compaction, presence TTLs, role-based revocation and field-sequenced conflict detection across offline reconnect and recovery paths.
- Building a collaborative architecture workspace with bound connectors, freehand drawing, multi-select, copy/paste, collaborative-safe undo/redo, styling, version history and editor/viewer sharing, plus a 25+ component system-design library; validated through 190+ PostgreSQL integration tests, Playwright visual/E2E tests, k6 load tests, CI/CD, and multi-replica failure-injection and recovery suites.

TECHNICAL SKILLS

Languages & Core: Go, C++, Python, TypeScript, JavaScript, SQL, Data Structures & Algorithms, System Design, Distributed Systems

Backend & Distributed Systems: FastAPI, Pydantic, Node.js, Express.js, Fastify, REST APIs, WebSockets, Socket.IO, Kafka, event-driven systems, idempotency, transactional outbox, concurrency control, distributed tracing

Data & AI Systems: PostgreSQL, Redis, MongoDB, MySQL, SQLite, Azure AI Search, Azure OpenAI, LangGraph, RAG, BM25/Hybrid Retrieval, FAISS, SentenceTransformers, Cross-Encoder Reranking, Embeddings, Retrieval Evaluation, deterministic replay

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions, self-hosted runners, Nginx, CI/CD, OpenTelemetry

Frontend & Testing: React.js, Next.js, Konva, Tailwind CSS, IndexedDB, Playwright, k6, integration testing, failure-injection testing, load testing

ACHIEVEMENTS

Competitive Programming: LeetCode Knight - peak rating 1923 (Top 5.6%), 1,500+ problems solved and a 550+ day Problem-of-the-Day streak; Codeforces Specialist - peak rating 1415, with a Global Rank of 818 in Codeforces Round 952.

Programs & Competitions: Selected for Amazon ML Summer School 2024; placed Top 3 in GCOS 2024 at IIT Kharagpur and secured a Top 5 finish in Overnite at IIT Kharagpur's Kshitij, a timed algorithmic programming contest.

Academic Honour: Awarded IIT Kharagpur's Shyamal Ghosh & Sunanda Ghosh Endowment Honour for 2021-22 academic excellence.